



GRIDLOCK

Event-Driven Congestion Intelligence for Bengaluru Traffic

Forecast → Quantify Impact → Prescribe Deployment → Learn

Team TrafficSolvers

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Flipkart Gridlock Hackathon 2.0 · Round 2

Built on the anonymized ASTraM incident dataset — 8,173 events × 46 fields

Events break the city — and we react blind

Rallies, festivals, sports, construction & sudden gatherings cause localized traffic breakdowns. Today the response is reactive and experience-driven.

① Impact is not quantified

No way to know in advance how bad an event will be, how long it will tie up the road, or which junctions it threatens.

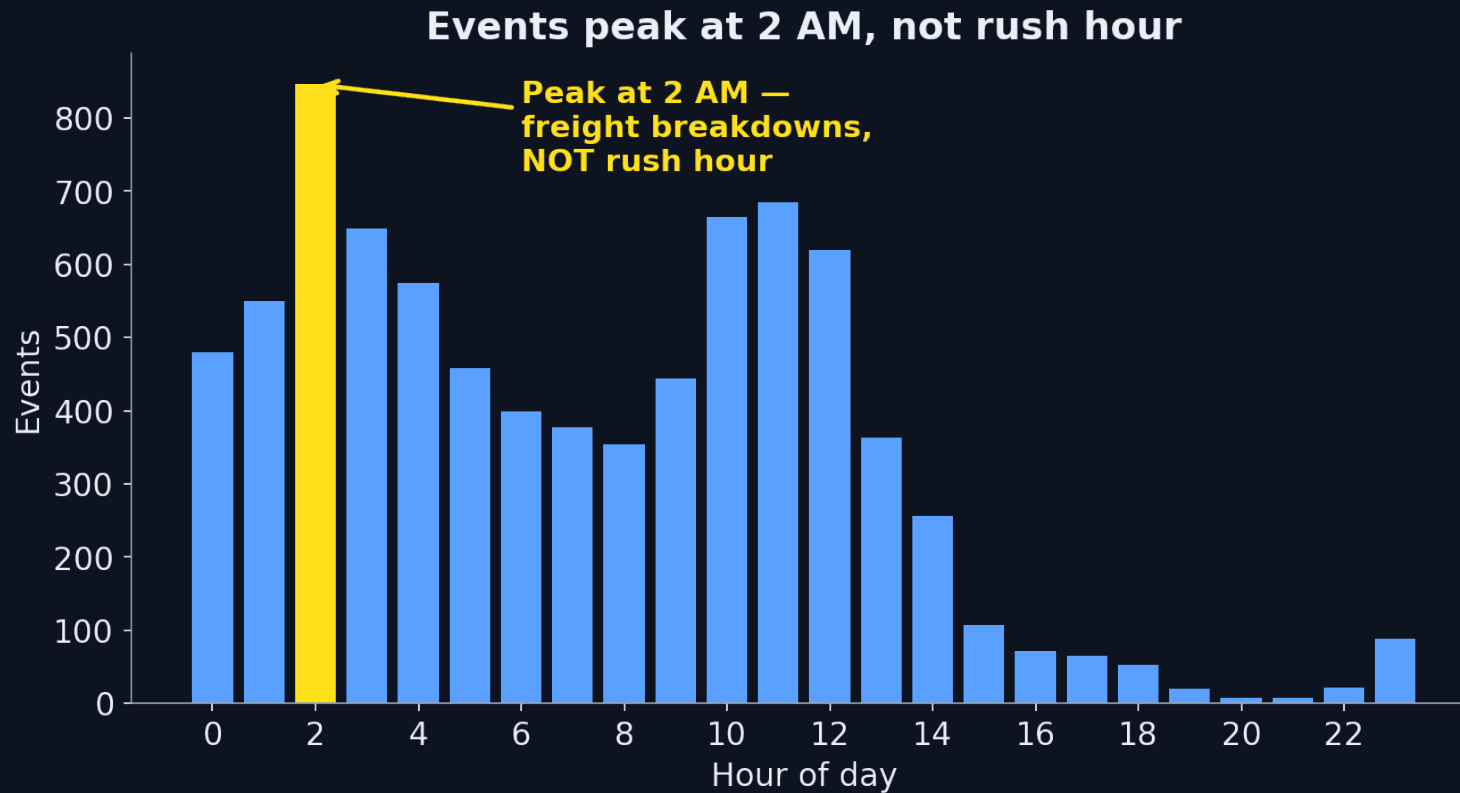
② Deployment is guesswork

Officers, barricades and diversions are assigned from experience — not from data, and not optimized against a finite resource pool.

③ No post-event learning

Every event is handled fresh. Mistakes are never fed back; the system never gets smarter after the fact.

This is an incident log — not congestion data



There is NO speed / volume / delay field.

So the target — “traffic impact” — does not exist in the data. It must be engineered.

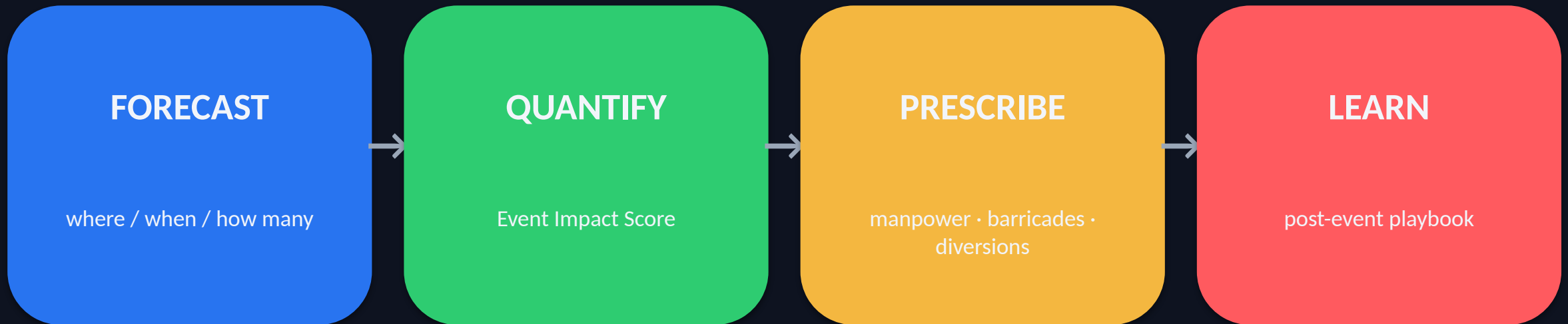
Events peak at 2 AM, not rush hour:

60% are vehicle breakdowns, driven by night-time freight movement. Teams that assume normal rush-hour patterns build the wrong model.

We turn this gap into our centrepiece.

Not prediction — a decision-support system

Most teams stop at “predict congestion.” We deliver the full operational loop:



Two pipelines — Planned (proactive scheduling) and Unplanned (real-time triage) — exactly matching the problem statement.

Four layers on one data foundation

L0 • DATA FOUNDATION

clean • leakage-safe features • H3 hex bins • EN+Kannada text

L1 • FORECAST

LightGBM-Poisson corridor counts • empirical-Bayes hotspots • Hawkes self-excitation

L2 • IMPACT (EIS)

clearance quantile regression + road-closure classifier → Event Impact Score

L3 • PRESCRIBE

manpower / barricades / diversions + city-wide officer optimization (ILP)

L4 • LEARN

predicted-vs-actual playbook • model-drift detection • retrain triggers

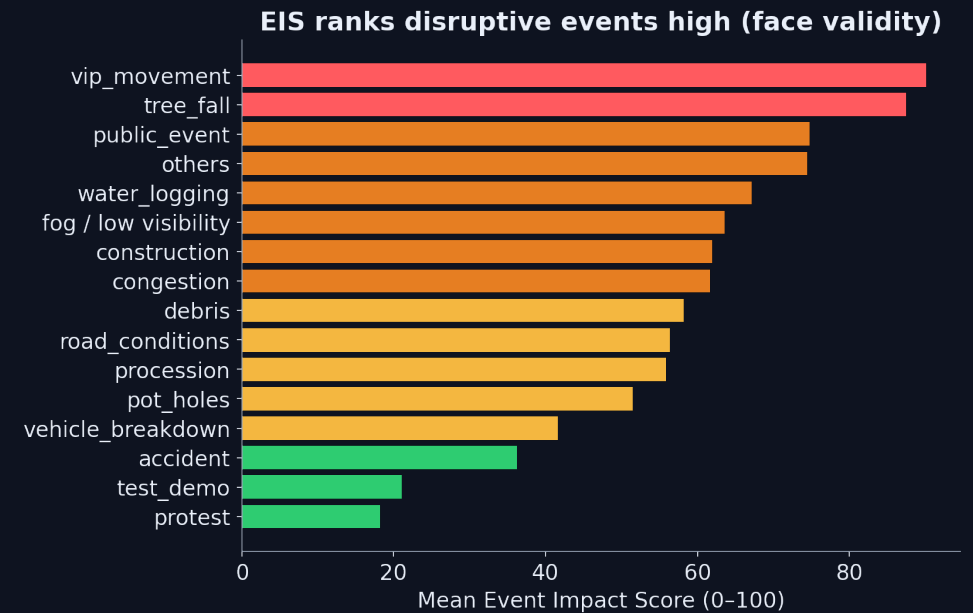
Event Impact Score — engineering the missing target

Framed as expected vehicle-hours of delay:

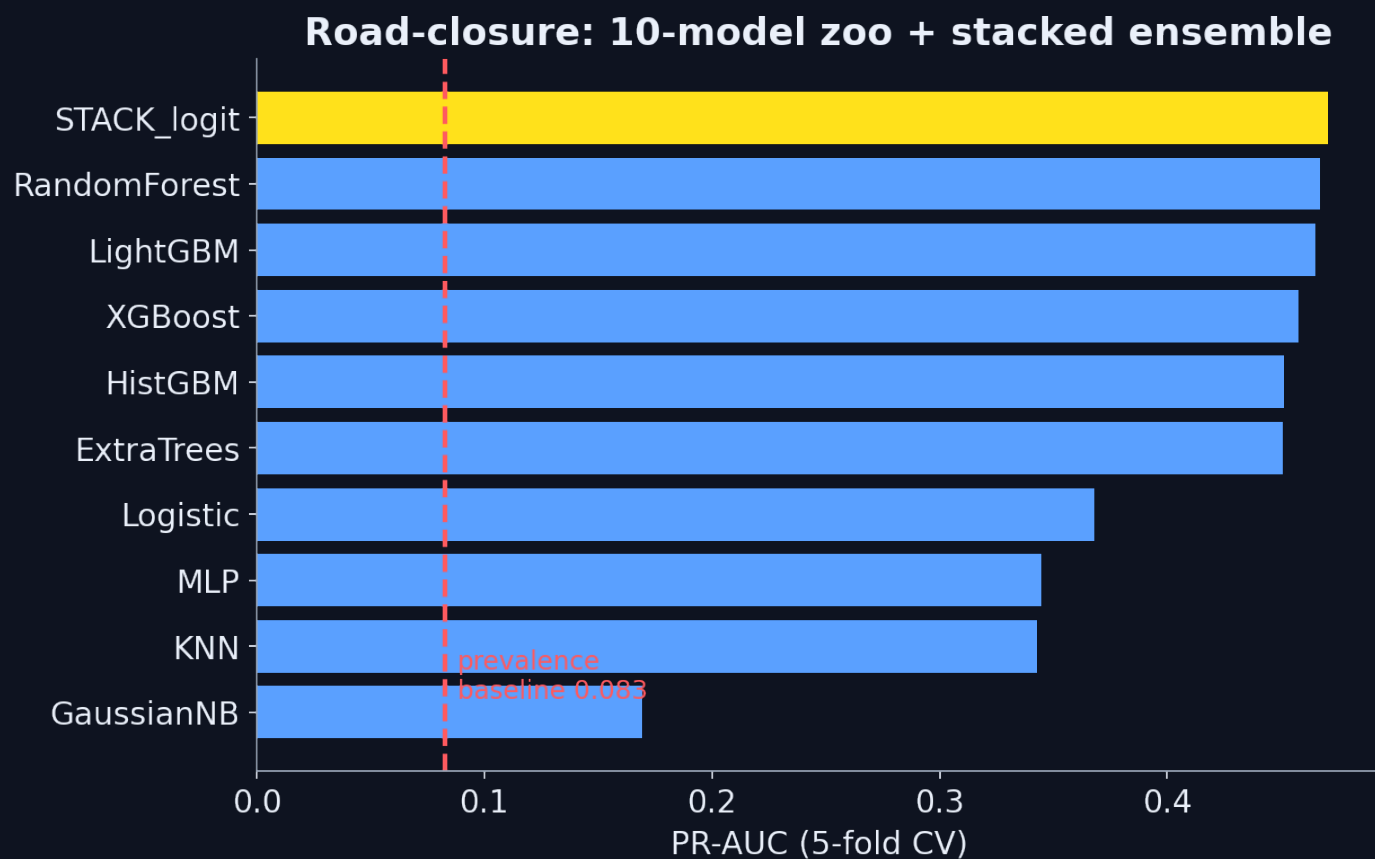
EIS $\approx$ Clearance $\times$ Spatial reach $\times$ Baseline flow $\times$ Closure severity

- Clearance time is predicted (leakage-safe), not observed
- Robust percentile-rank scaling $\rightarrow$ Low / Medium / High / Critical
- **VALIDATED against observed severity:**

Spearman 0.70 • AUC 0.89 flagging top-10% impact events



10-model zoo • k-fold CV • leakage hunt • Optuna



Headline results

Clearance time MAE 2.88 h (blended)

Road closure PR-AUC 0.471 • +469% vs baseline

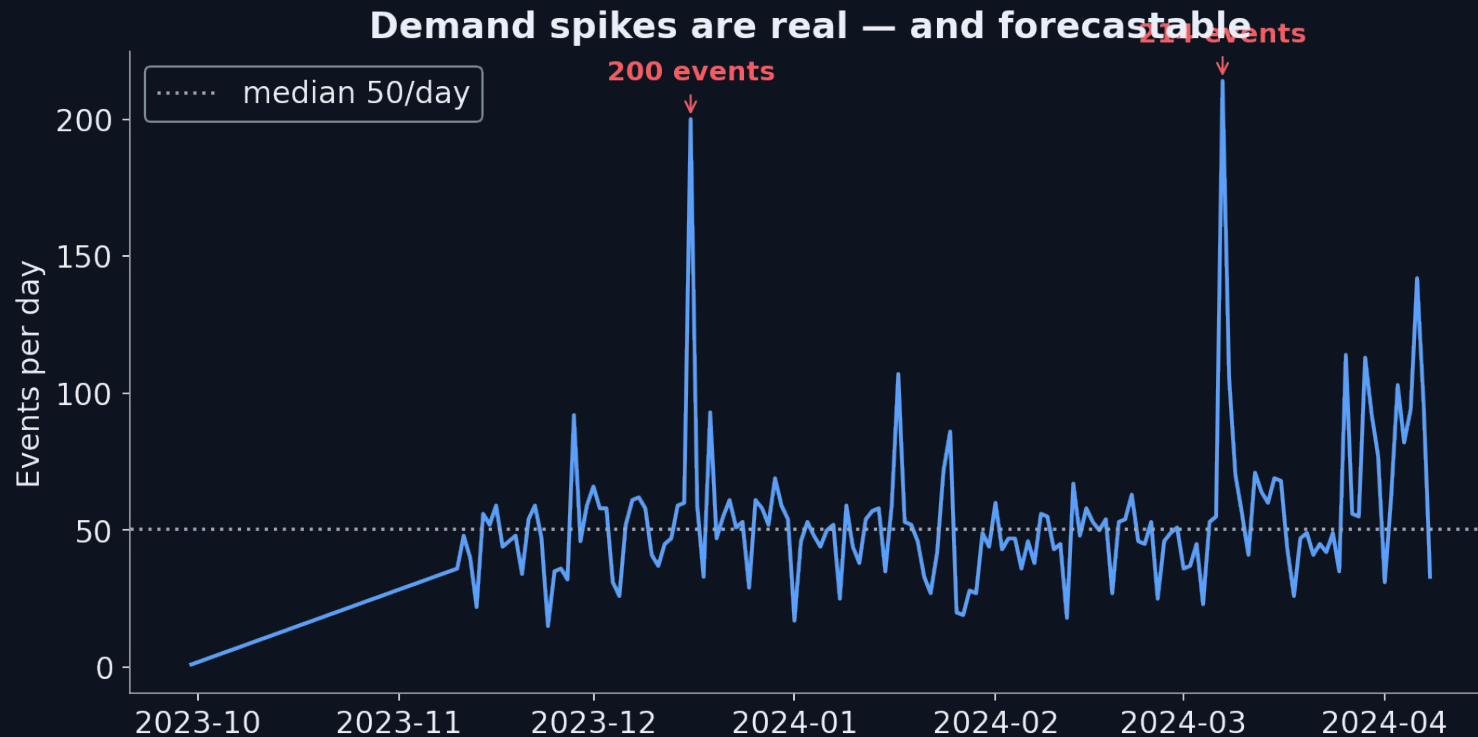
EIS validity Spearman 0.70 • AUC 0.89

Event forecast +16.7% over lag-7 baseline

Discipline that wins

- Time-split CV + spatial holdout (no leakage)
- Leakage hunt: dropped end-coords leaking @ AUC 0.976
- Priority shown ~location-circular — reported honestly
- Stacked ensemble + Optuna tuning

Where, when & how many events — ahead of time



LightGBM-Poisson

corridor × day event counts — 16.7% better than the lag-7 baseline.

Empirical-Bayes hotspots

recurring corridor × hour risk surface for pre-positioning.

Hawkes self-excitation

accidents cluster — branching ratio 0.31 — quantifying contagion.

The deliverable most teams never reach

Per-event plan

Officers, barricade units and a real road-network diversion (OSRM) — driven by the validated EIS tier + closure probability.

City-wide optimization

Finite officer pool allocated by ILP (PuLP) — two doctrines:

- Priority — cover highest-impact first (triage)
- Efficiency — maximize total mitigated impact

Surge day — 7 Mar 2024

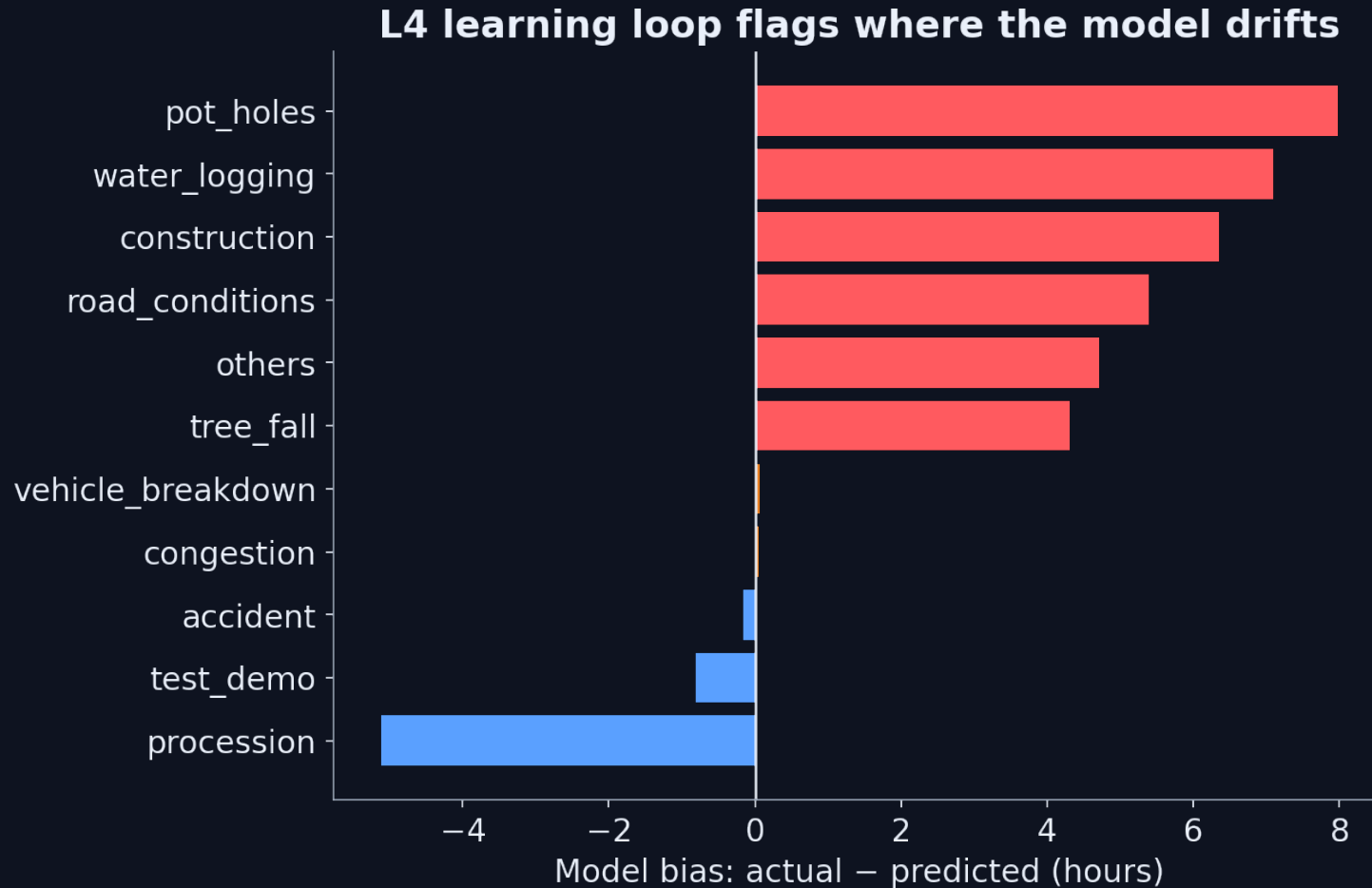
214 events in one day

≈ 1,096 officers needed

vs ~120 available → a quantified deficit

The system tells the commander exactly which events to staff first and how many more officers to escalate for — turning guesswork into math.

The post-event loop the brief explicitly asks for



After events close,

we compare leak-free predictions with reality and build a playbook keyed by cause × H3 hex.

It self-diagnoses:

the model under-predicts infrastructure events — potholes by +8.0 h, water-logging +7.1 h — and flags those cells for retraining.

Closes the “no post-event learning” gap most teams ignore.

A live decision-support dashboard

 Event Simulator

score any what-if event → EIS, clearance P50/P90, full deployment plan

 City Risk Map

all 8,173 events on a Bengaluru map, coloured by impact tier

 Surge-Day Replay

replay a real day, set an officer budget → live allocation + deficit

 Forecast & Hotspots

corridor × hour risk heatmap, Hawkes, the 2-AM explainer

 Learning Loop

predicted-vs-actual playbook + drift alerts

Live proof: the same breakdown scores EIS 17 at 2 AM vs 78 at 9 AM rush; a VIP movement scores 99.7 (Critical) with a real OSRM diversion drawn on the map.

Seven things the field won't have together

Correct reframe

incident log, not congestion data

Engineered + validated EIS

Spearman 0.70 — the missing target, proven

k-fold model zoo

10 models, stacking, Optuna — not one model

Prescription + ILP

we allocate scarce officers; others only predict

Post-event learning

L4 playbook + drift — the 3rd gap, solved

Real-ML simulator

runs the trained models, not random numbers

Honesty & rigor

leakage hunt, time-split CV, stated limits

Operational value — and what we'd add next

Operational value

- Quantified impact before the event, not after
- Optimal, defensible resource deployment
- Faster clearance via right-sized response
- A city that gets smarter after every event

Roadmap

- Plug in live flow (Google/TomTom) to calibrate EIS
- Spatio-temporal GNN forecasting · officer-shift scheduling

Honest limitations (we say so)

- No ground-truth flow → EIS validated vs proxies
- Manpower data sparse (1.6%) → optimization + rules, not over-claimed as learned
- 5 months of data → calendar priors mitigate seasonality

Stack

Python · LightGBM/XGBoost/CatBoost · scikit-learn · Optuna ·
H3 · PuLP · OSRM · Streamlit + pydeck

From reaction to anticipation.

Forecast the event · quantify the impact · deploy the optimal response · learn from every outcome.

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